Values of $q(x)$ are estimated as

$$q(40) = \frac{d(40)}{l(40)}$$

$$= \frac{102}{10645}$$

$$= 0.0095.$$

$$q(41) = \frac{d(41)}{l(41)}$$

$$= \frac{169}{10543}$$

$$= 0.0160.$$

Values of $p(x)$ are estimated from the corresponding values of $q(x)$ as

$$p(40) = 1 - q(40)$$

$$= 1 - 0.0095 = 0.9905$$

$$p(41) = 1 - q(41)$$

$$= 1 - 0.0160 = 0.9840$$

Values of $L(x)$ are estimated using its relationship with $l(x)$ and $d(x)$ as follows:

$$L(40) = \frac{l(40) + l(41)}{2}$$

$$= \frac{10645 + 10543}{2}$$

$$= 10,594$$

$$L(41) = l(41) - \frac{1}{2}d(41)$$

$$= 10,543 - \frac{1}{2} \times 169 = 10,458.5$$

$$= 10,459 \text{ (}Approx.\text{)}$$

The value of $T(41)$ is estimated from the given value of $T(40)$ and the estimated value of $L(40)$ from the relationship

$$T(41) = T(40) - L(40)$$

as $T(41) = 193820 - 10594$

$$= 1,83,226.$$

The life expectancy of the ***cohort*** at the age $x = 40$ and 41 years can be estimated using the relationship

as

$$e^0(x) = \frac{T(x)}{l(x)}$$

$$e^0(40) = \frac{1,93,820}{10,645} = 18.20$$

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